

5 (long)

Figure 5(a) shows an arrangement of beams of uniform cross section having flexural rigidity EI . The beams support a uniformly distributed load w per unit horizontal length.

- (a) Draw the shear and bending moment diagrams for the beam subject to this loading arrangement, noting all salient values. [10]
- (b) Determine the midspan displacement at point C in terms of w , L , E and I . [10]
- (c) A possible design for the beam using a welded triangular box section manufactured from 10 mm thick steel plate is shown in Fig. 5(b). If the steel plate has a yield strength of 275 MPa, what is the maximum value of bending moment that the section can sustain before first yield? [10]

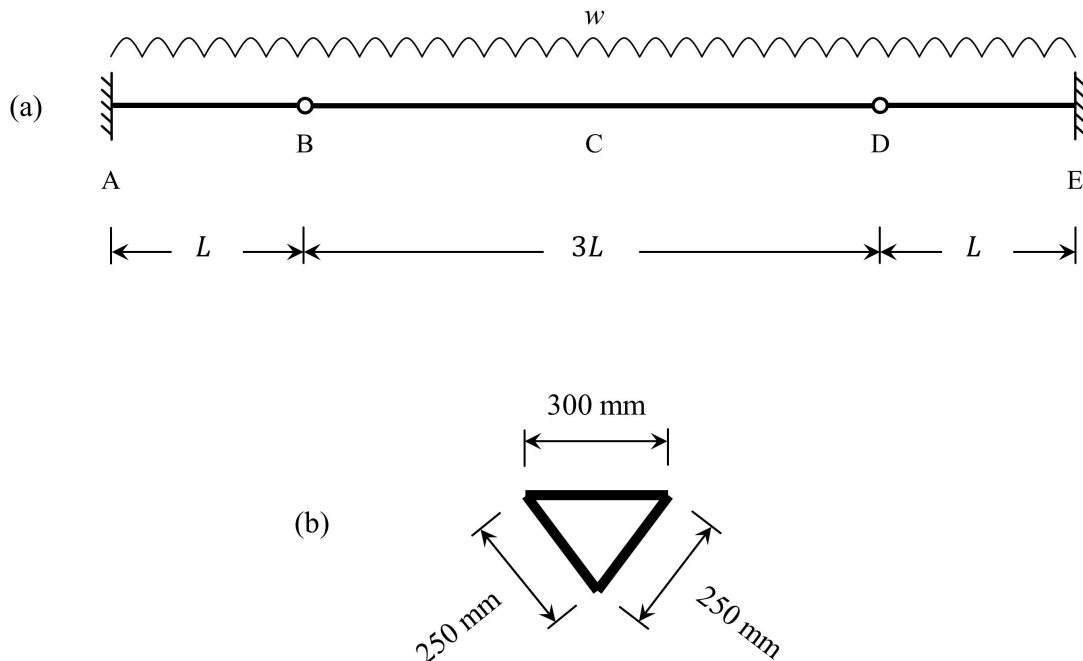


Fig. 5